

Hayatullah Ahmadzai

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Education

2015–2019: Ph.D. in Economics (focus: Agricultural and Development Economics), University of Nottingham, UK

2008–2011: M.S. Agricultural Economics, Purdue University, USA

2002–2005: B.S. Agronomy and Soil Sciences, Kabul University, Afghanistan

Primary Research Interests: Agricultural & Food Policy • Climate Change, Adaptation, & Agricultural Resilience • Agribusiness & Farm Economics • Food Security, Rural Livelihoods & Agrifood Systems • Applied Econometrics & Impact Evaluation

Academic and Research Appointments

06/2025–Present: External Research Fellow, Center for Research in Economic Development and International Trade (CREDIT), University of Nottingham, Nottingham, UK (remote, based in Durham, NC, USA).

06/2022–06/2025: Research Fellow, Center for Research in Economic Development and International Trade (CREDIT), University of Nottingham, Nottingham, UK.

04/2019–04/2022: Postdoctoral Research Fellow, International Center for Biosaline Agriculture, Dubai, UAE.

02/2012–09/2015: Senior Economist/Head of Monitoring, Evaluation & Learning (MEL), Department for International Development (DfID, now FCDO), Kabul, Afghanistan.

03/2011–09/2015: Assistant Professor (Teaching-focused, Part-time), Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan.

07/2011–02/2012: Economist and Value-Chain Development Specialist, FLAG International Afghanistan (FIA), Kabul, Afghanistan.

03/2006–10/2007: Teaching Assistant, Department of Agronomy and Soil Sciences, Kabul University, Kabul, Afghanistan.

Fellowships and Scholarly Appointments

2025–2027: MESA Global Academy Scholar, Middle East Studies Association (MESA), Washington, DC, USA (remote, based in Durham, NC, USA).

Awarded a one-year scholarship (2025–2026), competitively renewed for a second year (2026–2027); conducting policy-relevant research and engaging with scholars on Middle East and North Africa issues.

Publications

Peer-reviewed Journal Articles

Climate shocks, household food security and welfare in Afghanistan

Ahmadzai, H., and Morrissey, O. 2025. *Food Policy*, 134, 102910.

Crop Diversification and Farm Household Welfare during Conflict: Afghanistan 2011–2017

Ahmadzai, H., and Morrissey, O. 2025. *The Journal of Development Studies*, 61(3), 381–399.

Tests for market failures and overcoming transaction costs barriers to market participation in Afghanistan

Ahmadzai, H. 2024. *Journal of Social and Economic Development*.

Empirical Analysis of Impacts of Sand and Dust Storms on Agricultural Productivity in Mongolia

Ahmadzai, H., Malhotra, A., and Tutundjian, S. 2023. *PLOS ONE*, 18(2), e0269271.

The Impact of Sand and Dust Storms on Agriculture Productivity in Iraq

Ahmadzai, H. 2023. *Middle Eastern Development Journal*, 15(1), 50–65.

Policies for Sustainable Agriculture and Livelihood in Marginal Lands: A Review

Ahmadzai, H., Elouafi, I., and Tutundjian, S. 2022. *Sustainability*, 13(16), 8692.

Orphan Crops: A Best Fit for Dietary Enrichment and Diversification in Highly Deteriorated Marginal Environments

Talabi, A., Vikram, P., Thushar, S., Hifzurrahman, R., Ahmadzai, H., Nhamo, N., Shahid, M., and Singh, R. 2022. *Frontiers in Plant Science*, 13, 839704.

Hope for Change: Is Diversifying Production Portfolios an Ideal Strategy to Boost Farming Efficiency in Afghanistan?

Ahmadzai, H. 2021. *Progress in Development Studies*, 21(3), 1–25.

Development of Quinoa Value Chain to Improve Food and Nutritional Security in Rural Communities in Rehamna, Morocco: Lessons Learned and Perspectives

Hirich, A., Rafik, S., Rahmani, M., Fetouab, A., Filali, K., Ahmadzai, H., et al. 2021. *Plants*, 10(2), 301.

How is Off-farm Income Linked to On-farm Diversification? Evidence from Afghanistan

Ahmadzai, H. 2020. *Studies in Agricultural Economics*, 122(1), 1–12.

Trends in Quinoa Adoption in Marginal Areas, Economic Viability, and Policy Outlook

Ahmadzai, H. 2020. *Journal of Agribusiness and Rural Development*, 57(3), 235–247.

Book Chapters

Marginal Areas and Indigenous People Priorities for Research and Action

Sayed, A., and Ahmadzai, H., et al. 2023. In *Science and Innovations for Food Systems Transformation*. Springer.

Impacts of Sand and Dust Storms on Agricultural Production

Middleton, N., Ahmadzai, H., and Ayurzana, E. 2023. In *Sand and Dust Storms: A Guide to Mitigation, Adaptation, Policy and Risk Management Measures in Agriculture*.

From Desert Farm to Fork: Value Chain Development for Innovative Salicornia-based Food Products—A Case Study in the United Arab Emirates

Lyra, D., Lampakis, E., et al., Ahmadzai, H., et al. 2021. In *Saline Futures: State-of-the-Art in Saline Agriculture*.

Policy Papers & Technical Reports

Marginal Lands: Potential for Agricultural Development, Food Security and Poverty Reduction

Ahmadzai, H., Tutundjian, S., et al. 2022. *SOLAW21 Technical Background Report*, FAO.

Agri-tech: A Catalyst for Treated Wastewater Deployment in High Efficiency Food Production Systems

Ammar, K., Fatnassi, H., and Ahmadzai, H., et al. 2020. *International Center for Biosaline Agriculture (ICBA)*.

Marginal Areas and Indigenous People: Priorities for Research and Action

Sayed, A., and Ahmadzai, H., et al. 2022. *Food Systems Summit Brief*.

Addressing Sand and Dust Storms in Northeast Asia (China and Mongolia) and Western Asia (Iran, Iraq, and Kuwait) and North Africa (Algeria): An Agricultural Perspective

Ahmadzai, H. 2022. *FAO and ICBA*.

Baseline Survey Report: Improving Agricultural Resilience to Salinity (RESADE)

Ahmadzai, H. 2020. International Center for Biosaline Agriculture (ICBA), IFAD, BADEA.

Working Papers

Security, Infrastructure, and Gender Gap in Rural Labor Markets: Evidence from Afghanistan's Repeated Cross-Sectional Survey

Ahmadzai, H.

CREDIT Research Paper; work in progress

Research Grants and Projects

2022–2024: Institute of International Education (IIE) and CARA (Council for At-Risk Academics), IIE-SRF Afghanistan Emergency Initiative Fund. **Principal Investigator.** £86,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Catalyzing Investments and Actions to Enhance Resilience Against Sand and Dust Storms in Agriculture. **Principal Investigator.** \$150,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Climate-Resilient and Gender-Inclusive Cropping Systems in Climate-Vulnerable Areas in Asia and the Pacific. **Co-Principal Investigator** with R.K. Singh (ICBA). \$178,441.

2021: Ministry of Climate Change and Environment, UAE, Developing National Policy to Utilize Treated Sewage Effluent (TSE) and Biosolids in Controlled Environment Agriculture in the UAE. **Co-Principal Investigator** with Khalil Ammar (ICBA). \$100,000.

2020: International Center for Biosaline Agriculture (ICBA), Adoption and Impact of Quinoa Among Farmers in Marginal Areas of the MENA Region. **Principal Investigator.** \$68,488.

2020: University of Wollongong in Dubai and ICBA, Resilient Agrifood Dynamism Through Evidence-Based Policies (READY). **Co-Principal Investigator** with Ioannis Manikas (University of Wollongong) and Khalil Ammar (ICBA). \$1,500,000.

2019: IFAD and BADEA, Improving Agricultural Resilience to Salinity and Climate Change through Development of Pro-Poor Technologies and Management Strategies (RESADE). **Co-Principal Investigator** with R.K. Singh (ICBA). \$7,700,000.

2019: International Center for Biosaline Agriculture (ICBA), EXPO Live Phase II: Value Chain Development for Innovative Halophyte-Based Food Systems. **Co-Principal Investigator** with Dionysia Angeliki Lyra (ICBA). \$500,000.

Courses Taught

Assistant Professor (Part-time), Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan.

2011–2015: AgEcEx731: Agricultural Production Economics (undergraduate, spring terms)

2011–2015: AgEcEx836: Agricultural Finance (undergraduate, fall terms)

2012–2014: AgEcEx630: Introductory Statistics & Econometrics (undergraduate, spring terms)

2011–2015: AgEcEx735: Research Methods & Projects (undergraduate; seminar-based teaching and student research supervision)

Selected Presentations

Conference Presentations

2026: “Climate Shocks, Household Food Security, and Welfare in Afghanistan.” Stanford University (FSE, Center for South Asia, Abbasi Program in Islamic Studies), April 22, 2026. Event | Slides | Video

2023: “Crop Diversification and Farm Household Welfare during Conflict: Afghanistan 2011–2017.” 6th MaDE Workshop on Emerging and Developing Countries, Loughborough Business School (online), June 5–6, 2023. Slides

2021: “Assessing the Impact of Sand and Dust Storms on Agriculture.” Food and Agriculture Organization (FAO), Bangkok (online), April 7–11, 2021. Slides

2020: “Salinization in Marginal Agricultural Areas: Implications for Global Food Security and Poverty.” International Center for Biosaline Agriculture (ICBA), Dubai, UAE, June 27–28, 2020. Slides

2018: “Tests for Market Failures and Overcoming Transaction Costs in Afghanistan.” Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, September 8–12, 2018. Slides

2017: “Status, Patterns, and Microeconomic Drivers of the Extent of Diversity in Crop Production: Evidence from Afghanistan.” Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 15–20, 2017. Slides

2016: “Crop Diversification and Technical Efficiency: Evidence from Afghanistan.” Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 12–16, 2016. Slides

Invited Talks

2021: “Challenge and Opportunity of Quinoa for Food and Nutrition Security.” Panellist with Warren T.K. Lee, chaired by Bo Zhou. FAO Regional Office for Asia and the Pacific (FAO-RAP), Bangkok, Thailand, November 7–8.

2021: “The Economics of Quinoa Production: Opportunity for Marginal Farmers.” Panellist with Dr. José Graziano da Silva. Global Forum on Innovations for Marginal Environments (GFIME), Dubai, UAE, November 20–21.

Awards

2021: Co-recipient, Khalifa International Award for Date Palm and Agricultural Innovation (KIADPAI), awarded as Co-PI of the Project: “From Desert Farm to Fork: Value Chain Development for Innovative Halophyte-Based Food Products.”

2015: PhD Scholarship Award, University of Nottingham and Islamic Development Bank (IsDB).

2007: Graduate Scholarship Award, Purdue University and USAID-funded Afghan Merit Scholar Program (AMSP).

Service and Professional Activities

2019–present: External Research Fellow, Centre for Research in Economic Development and International Trade (CREDIT), University of Nottingham, UK.

2019–present: Member, Agribusiness Sub-network, Association of Agricultural Research Institutions in the Near East and North Africa (AARINENA).

2019–present: Peer Reviewer for *The Journal of Development Studies*; *Food Policy*; *Sustainability* (MDPI); *PLOS ONE*; *Review of Development Economics*; *Humanities and Social Sciences Communications*; *Agriculture* (MDPI); *Cogent Economics and Finance*; *Journal of Social and Economic Development*.

Media and Outreach

2022: “How COVID-19 Crisis Undermined MENA States’ Food Security.” *Arab News*, November 20, 2022.

2021: “Agriculture in Marginal Lands.” *Scholarly Community Encyclopedia*, September 2021.

2021: “New Study Reviews Status of Marginal Lands Worldwide.” ICBA News Article, August 29, 2021.

2021: “World’s Indigenous Peoples to Bear Brunt of Climate Change, Warns UN Paper.” ICBA News Article, April 25, 2021.

2020: “ICBA’s Quinoa Varieties Show Promise in Morocco.” ICBA News Article, April 15, 2020.

Professional Development

2020: Responsible Conduct in Research: Research, Ethics, and Compliance. Collaborative Institutional Training Initiative (CITI). Completed online, March 24–April 4.

2013: Leadership and Governance of Development Programs and Projects. Crown Agents International. Dubai, United Arab Emirates, November 11–15.

2012: Management of Agriculture and Rural Development Projects. Asian Institute of Technology (AIT). Bangkok, Thailand, November 21–December 1.

2008: Academic English and Cross-Cultural Training Program. International Centre for Intercultural Communication (ICIC). IUPUI, Indianapolis, IN, USA, October 29, 2007–May 30, 2008.

Skills

- **Economic & Policy Analysis:** Impact evaluation, program evaluation, benefit-cost analysis, value-for-money analysis, cost-effectiveness analysis, project appraisal, feasibility studies, policy analysis and design, welfare and distributional analysis, economic valuation, decision analysis.
- **Quantitative Methods:** Applied econometrics, causal inference, quasi-experimental methods (IV, DiD, RDD, matching, discrete choice, panel data), experimental methods (RCTs, field experiments), survey design and sampling, household survey design and implementation, mixed-methods research.
- **Project and Grant Management:** Grant writing and proposal development, multi-partner project management, interdisciplinary collaboration, stakeholder engagement, capacity building and mentoring, technical and policy reporting, scientific writing, theory of change development, M&E framework design.
- **Software and Tools:** Stata, R, Python (Pandas, NumPy, Statsmodels, Seaborn, Matplotlib), Power BI, QGIS, Excel, Survey Solutions, Qualtrics, LaTeX, Markdown.

Linguistic

- English (fluent)
- Pashto (native)
- Dari/Persian (fluent)
- Arabic (conversational)